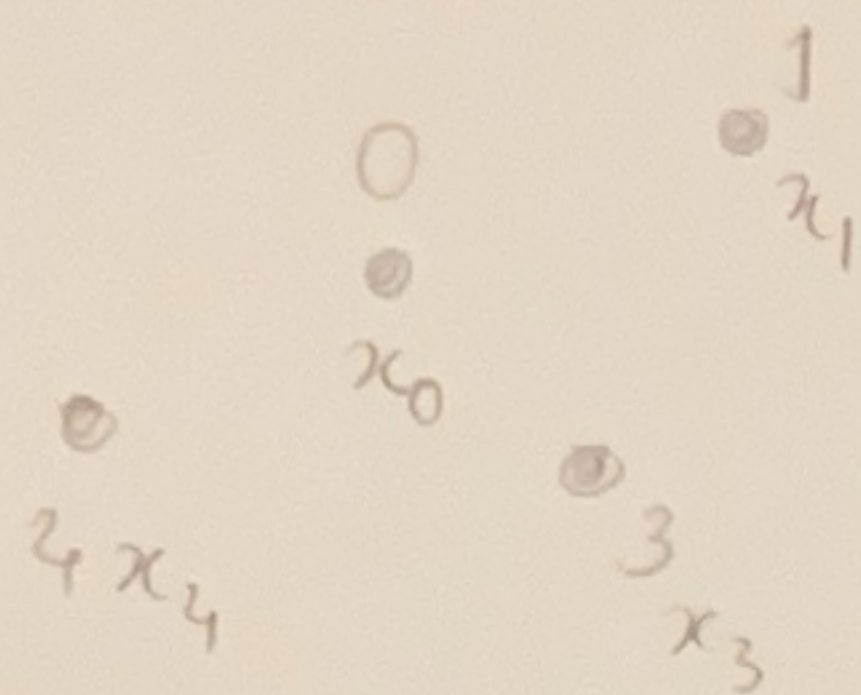


one graph has 4 agents.

node 0 $\rightarrow$ target model 1 $\rightarrow$ car model 2 $\rightarrow$ truck model 3 pedestrian



each Node i

$i \in \{0, 1, 2, 3, 4\}$

$\rightarrow$ has feature $x_i =$

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{15 \times 1}$$

① encoder (linear layers) $\rightarrow$ so to every node

$$x_0 \in \mathbb{R}^{15} \rightarrow \mathbb{R}^{64}$$

$$x_1 \in \mathbb{R}^{15} \rightarrow \mathbb{R}^{64}$$

$\rightarrow$ apply
linear
layers.

$$x_4 \in \mathbb{R}^{15} \rightarrow \mathbb{R}^{64}$$

$$x_0 \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{15} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{bmatrix}_{64} \xrightarrow{\text{Relu}} \begin{bmatrix} 0 \\ 0 \\ \vdots \end{bmatrix}_{64} \rightarrow \mathbb{R}^{64} \quad h_1^{(0)}$$

$$x_1 \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{15} \rightarrow \boxed{\text{linear block}} \rightarrow \mathbb{R}^{64} \quad h_2^{(0)}$$

so for each Node x_i will have a output

after encoder as $h_i^{(0)}$ $\rightarrow$ cause GNN need

* general equation

$$\left[h_i^{(0)} = \text{Relu} (w_{\text{enc}} x_i + b) \right] \quad \text{Per node}$$

compute encoder output $h_i^{(0)}$

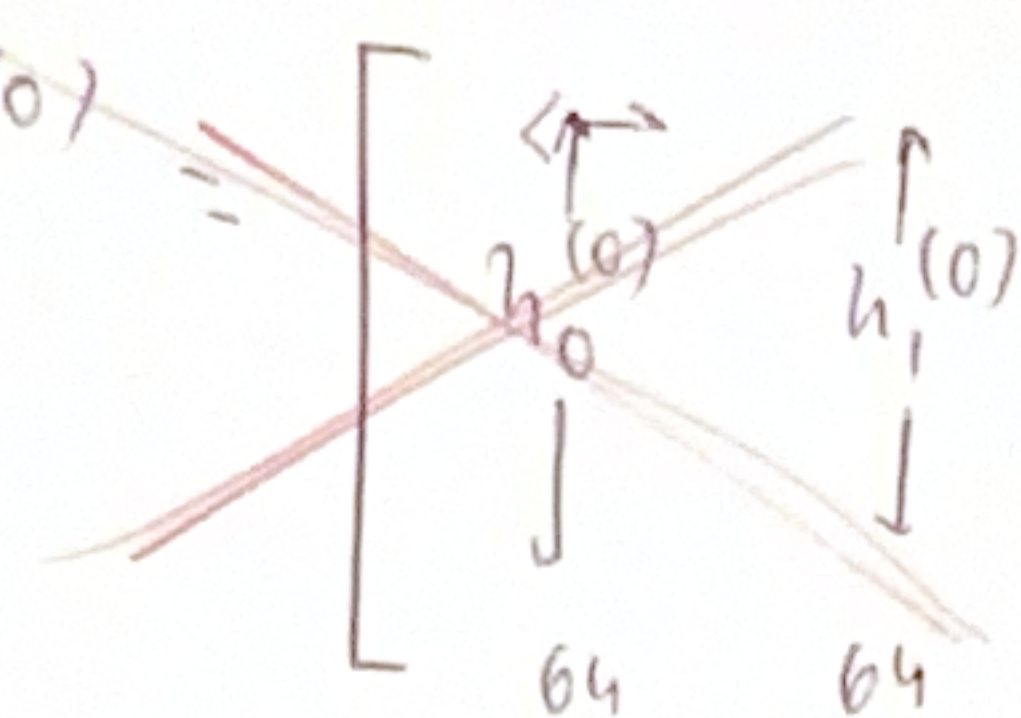
$$h_1^{(0)} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{64 \times 1} \quad \text{Node 1}$$

$$h_2^{(0)} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{64 \times 1} \quad \text{Node 2}$$

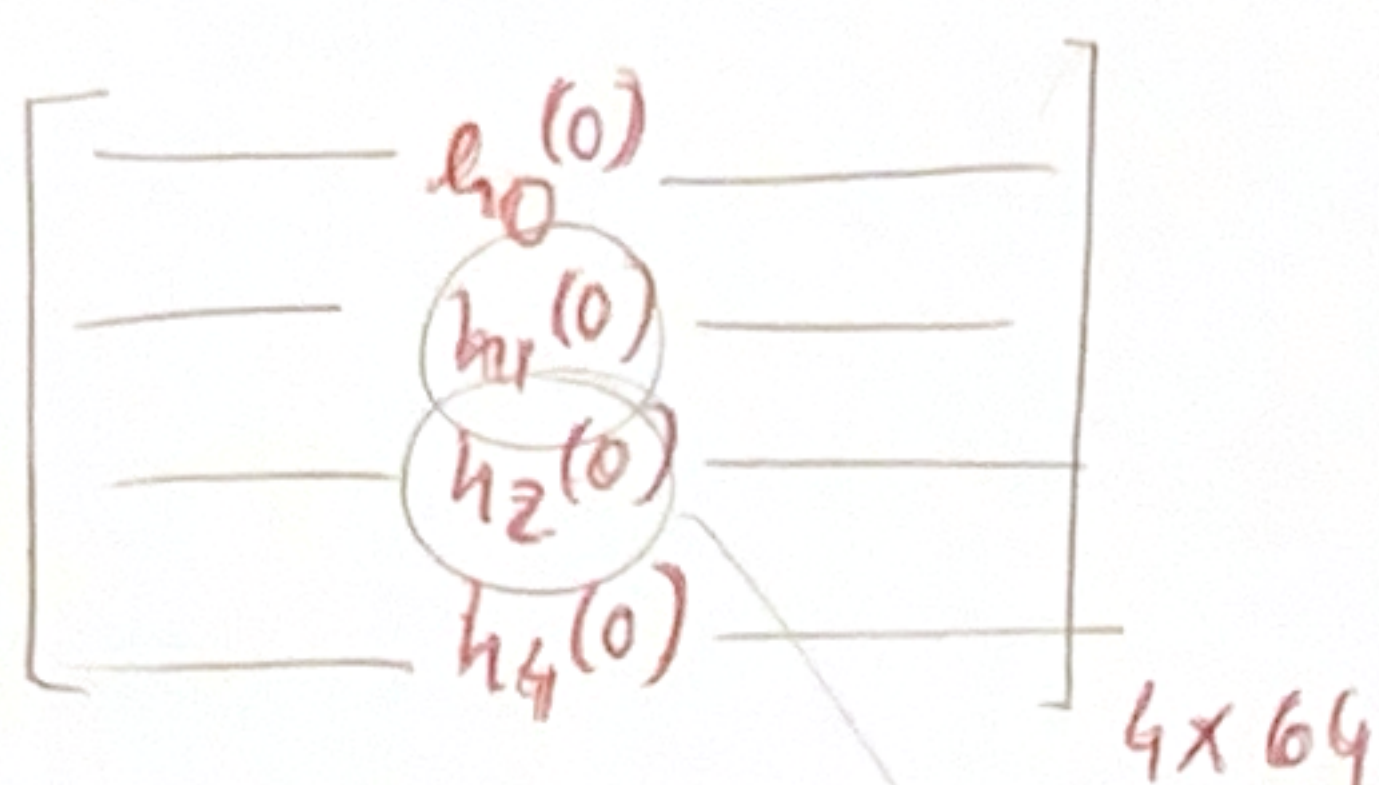
$$h_3^{(0)} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{64 \times 1} \quad \text{Node 3}$$

$$h_4^{(0)} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{64 \times 1} \quad \text{Node 4}$$

$$\rightarrow \text{concatenate } H^{(0)} = \begin{bmatrix} h_1^{(0)} & h_2^{(0)} & h_3^{(0)} & h_4^{(0)} \end{bmatrix}$$

so $H^{(0)} =$ 

at this point
there's still no
graph interaction

$H^{(0)} =$  4×64

② First GINE layer $\rightarrow$ suppose graph has edges.

- $1 \rightarrow 0$ $\rightarrow$ For target Node $i=0$

- $2 \rightarrow 0$ $N^{(0)} = \{1, 2\} \rightarrow$ incoming Neighbors.

- $0 \rightarrow 1$

- $0 \rightarrow 2$

- $2 \rightarrow 3$

- $3 \rightarrow 2$

so GINE builds messages

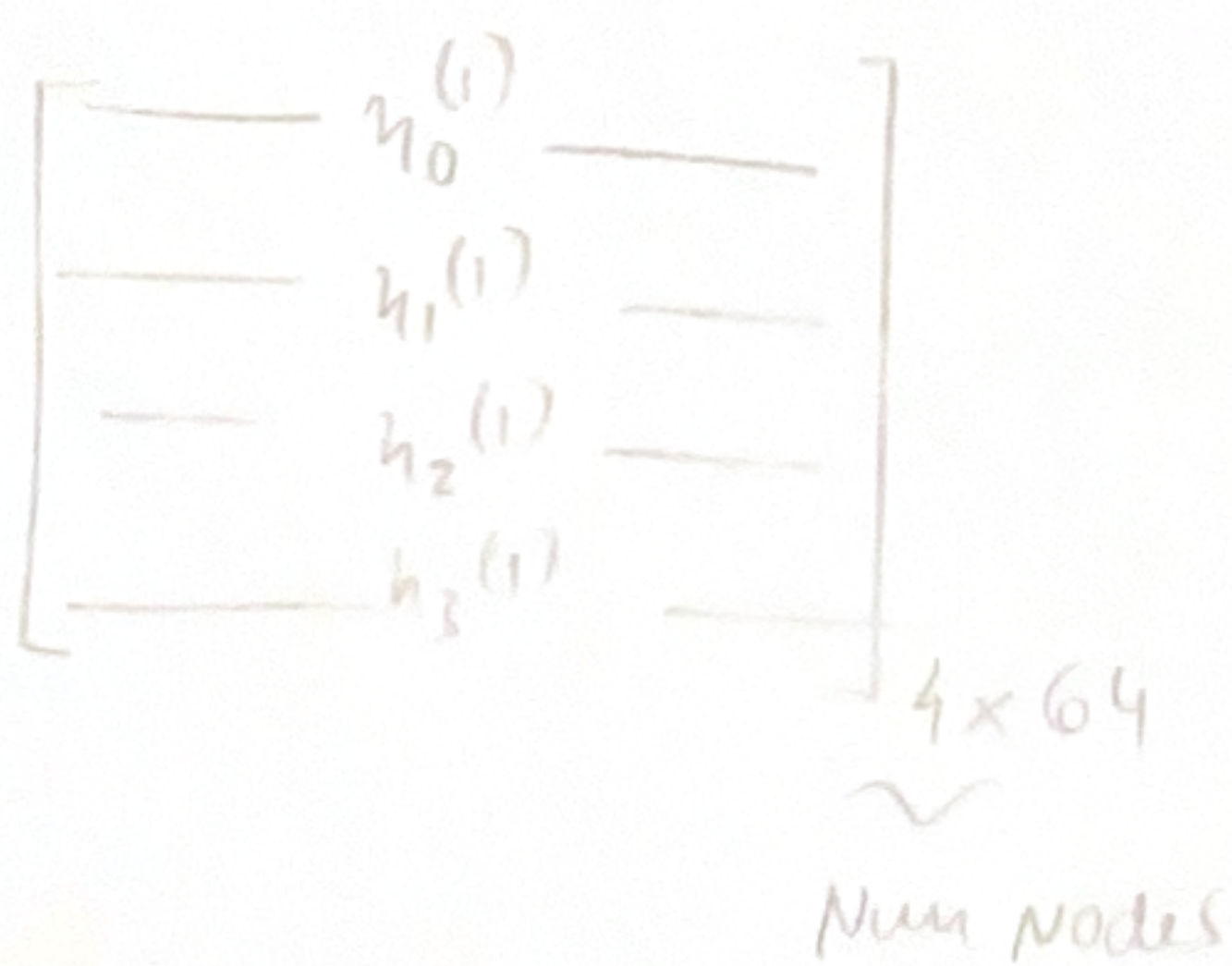
$\rightarrow m_{1 \rightarrow 0}^{(0)} = \text{ReLU}(h_1^{(0)} + w_e e_{10})$

$\rightarrow m_{2 \rightarrow 0}^{(0)} = \text{ReLU}(h_2^{(0)} + w_e e_{20})$

then aggregate in $M_0^{(0)} = m_{1 \rightarrow 0}^{(0)} + m_{2 \rightarrow 0}^{(0)}$ and

then update $\left[\begin{array}{l} \underbrace{h_0^{(1)}}_{\text{Node 0}} = \text{MLP}_1 \left((1 + \epsilon) h_0^{(0)} + M_0^{(0)} \right) \end{array} \right] \rightarrow$ update node 0
only, info neighbor

now do this for every Node in $\{0, 1, 2, 3, 4\}$.

so $H^{(1)} \in \mathbb{R}^{4 \times 64}$ $\rightarrow$ $H^{(1)} =$  4×64
after update
Num nodes

③ 2nd GIN Layer

Now Node i (2 per) may already contain info from nodes 3 of 1st layer

So apply same thing, again for node $i=0$

↳ check what are edges and incoming neighbors

$$\left. \begin{array}{l} 1 \rightarrow 0 \\ 2 \rightarrow 0 \\ 3 \rightarrow 0 \end{array} \right\} N(0) = \{1, 2, 3\}$$

$$\left. \begin{array}{l} 0 \rightarrow 1 \\ 0 \rightarrow 2 \\ 0 \rightarrow 3 \end{array} \right\} N(1) = \{0, 2, 3\}$$

$$2 \rightarrow 3$$

$$3 \rightarrow 2$$

GIN does message passing $m_{j \rightarrow i}$

$$m_{1 \rightarrow 0} = \text{Relu} \left(z_1^{(1)} + w_e e_{10} \right)$$

$$m_{2 \rightarrow 0} = \text{Relu} \left(z_2^{(1)} + w_e e_{20} \right)$$

$$m_{3 \rightarrow 0} = \text{Relu} \left(z_3^{(1)} + w_e e_{30} \right)$$

$$\text{Aggregate } M_0^{(1)} = m_{1 \rightarrow 0} + m_{2 \rightarrow 0} + m_{3 \rightarrow 0}$$

$$\text{or } M_2^{(1)} = m_{3 \rightarrow 2} + m_{0 \rightarrow 2}$$

so each node will get its own $M_i^{(L)}$

now update using

$$\left[\begin{array}{l} \text{Node 0} \\ \text{2° update} \end{array} \right] \begin{array}{l} \text{Prev} \\ \uparrow \\ h_0^{(2)} = \text{MLP}_2 \left((1+\epsilon) h_0^{(1)} + M_0^{(1)} \right) \end{array} \Rightarrow H^{(1)} \in \mathbb{R}^{4 \times 64}$$

$$\begin{bmatrix} h_0^{(1)} \\ h_1^{(1)} \\ h_2^{(1)} \\ h_3^{(1)} \end{bmatrix}$$

Nodes $\times$ 64

④ For prediction take only $h_0^{(1)} \equiv \text{TARGET}$

⑤ DECODER $\rightarrow 64 \rightarrow 128 \rightarrow 256 (12 \times 2) (x_1, x_2, \dots)$

each row $\equiv$ Node
 $\hat{Y} \in \mathbb{R}^{12 \times 2}$